LAST NAME: _____

Practice Quiz 08A — MATH 2055 — Fall 2026

Practice for Quiz 08

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

Write $[x]_5$ for the congruence class of the integer x modulo 5, and write $\mathbf{Z}_5$ for the set of these classes. Thus $[x]_5 = [y]_5$ if and only if $x - y = 5k$ for some integer k.

1. Consider the proposed rule

$$f : \mathbf{Z}_5 \longrightarrow \mathbf{Z}_5, \qquad f([x]_5) = [x+3]_5.$$

   (a) Prove that f is well defined.

   (b) Prove that f is onto and determine its inverse.